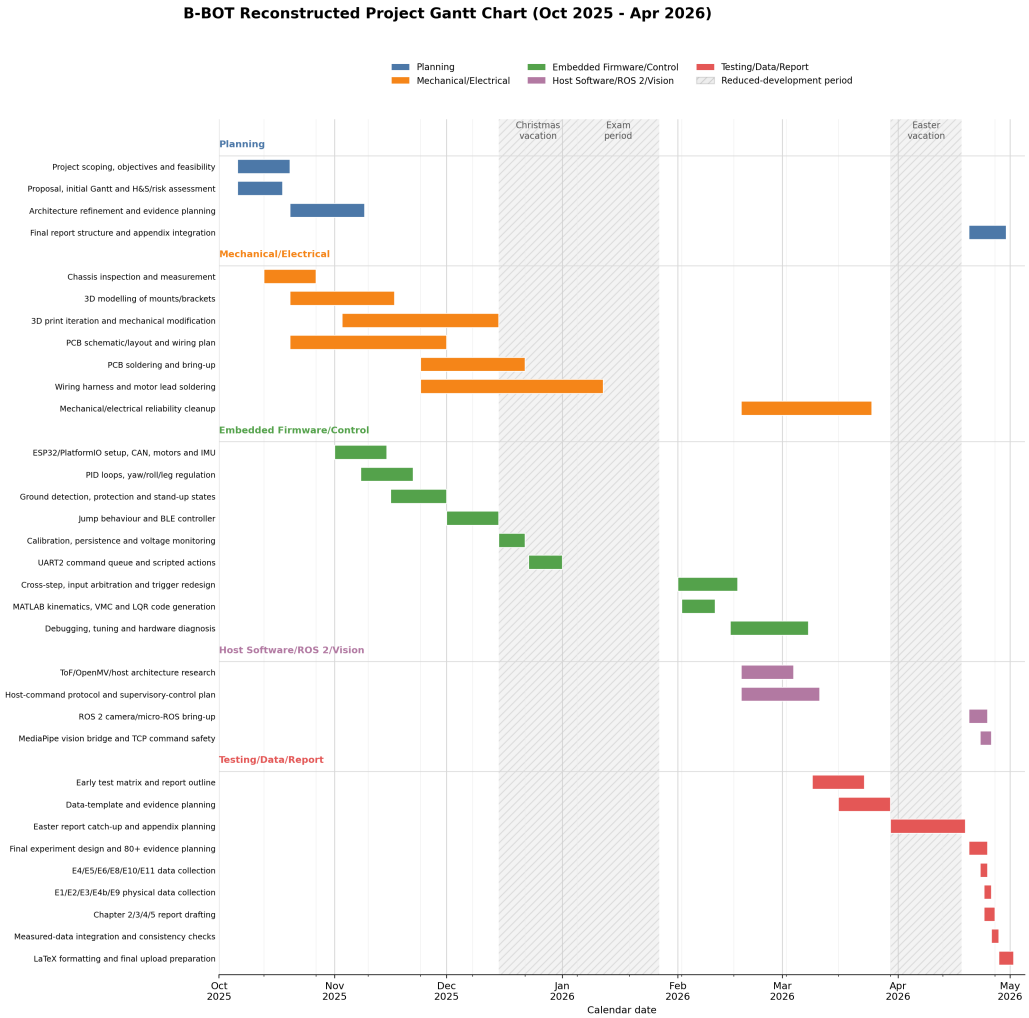
Appendix B: Project Plan, Gantt Chart and Weekly Activity Log

Generated companion PDF for marker-readable appendix review. The editable source files remain the authoritative Office/Markdown evidence.

Source Evidence

Evidence item	Location
Filled planning spreadsheet	Report/appendices/Project_management/Appendix_B_Project_Plan_Gantt/Project Planning Template - B-BOT filled.xlsx
Reconstructed Gantt and weekly log source	Report/planning/Project_Management_Gantt_and_Weekly_Log.md
Rendered Gantt PDF	Report/figures/project_management_gantt.pdf
Rendered Gantt PNG	Report/figures/project_management_gantt.png

Rendered Gantt Preview



Phase Summary

Phase	Approximate period	Evidence / output
Planning and management	Oct 2025 and Apr 2026	Proposal, objectives, risk/H&S records, reconstructed final schedule and appendix planning.
Mechanical and electrical build	Oct 2025 - Mar 2026	3D modelling, print iteration, PCB/wiring work, harness construction, motor soldering and integration cleanup.
Embedded firmware and control	Nov 2025 - Feb 2026	ESP32 PlatformIO firmware, CAN motor feedback, IMU processing, PID/LQR/VMC balance and safety states.
Host software, ROS 2 and vision	Feb 2026 - Apr 2026	Host architecture decision, micro-ROS camera bring-up, MediaPipe bridge and supervisory TCP command safety.

Phase	Approximate period	Evidence / output
Testing, data and report	Mar 2026 - Apr 2026	Experiment matrix, measured data package, generated figures, LaTeX report and appendix integration.

Reduced-Development Periods

Period	Reason	Project impact
2025-12-15 to 2026-01-05	Christmas vacation	Reduced lab time; mainly light firmware cleanup, documentation, calibration notes and command-system planning
2026-01-05 to 2026-01-27	Exam period	Minimal project development; focus shifted to revision and exams
2026-03-30 to 2026-04-18	Easter vacation	Lower formal contact/lab rhythm; used for catch-up, report planning, hardware cleanup and software architecture consolidation

Weekly Activity Log

Week	Dates	Main focus	Activities and outputs	Notes / status
1	2025-10-06 to 2025-10-12	Project initiation	Defined the project direction around an ESP32 wheel-legged self-balancing robot; reviewed initial hardware capability; drafted early aims, feasibility and project risks.	Reconstructed from project context; aligns with proposal/H&S timing.
2	2025-10-13 to 2025-10-19	Proposal, risk and architecture	Prepared proposal/risk material; identified hazards from unstable balancing, high-torque motors, battery power and wireless commands; started system-level architecture planning.	H&S/risk work linked to Canvas deadline period.
3	2025-10-20 to 2025-10-26	Mechanical/electrical planning	Measured chassis and components; began 3D modelling of mounts/brackets; sketched PCB/wiring plan; reviewed motor, IMU and CAN integration requirements.	Includes physical design work not fully reflected in git.
4	2025-10-27 to 2025-11-02	Build preparation	Continued CAD and print planning; started PCB schematic/layout iteration; prepared ESP32/PlatformIO project structure; began checking motor wiring and connector requirements.	Bridges planning to first firmware implementation.
5	2025-11-03 to 2025-11-09	Hardware foundation firmware	Set up ESP32 + PlatformIO + Arduino framework; implemented CAN bus communication; initialised six motors; integrated MPU6050 DMP; created motor send and CAN receive tasks.	Reflected in `Progress.md` 2025/11/1-11/7.
6	2025-11-10 to 2025-11-16	PID and manual control foundations	Implemented cascade PID framework; added roll, leg length, leg angle and yaw PID loops; added gravity compensation; continued mechanical/electrical checks.	Reflected in `Progress.md` 2025/11/8-11/15.
7	2025-11-17 to 2025-11-23	Ground detection and landing behaviour	Added support-force calculation, touch detection, anti-bounce logic, airborne wheel disable and landing cushioning state machine; reset position target on landing.	Reflected in `Progress.md` 2025/11/16-11/22.
8	2025-11-24 to 2025-11-30	Protection, stand-up and wiring	Added leg-angle and pitch protection; implemented split-leg stand-up state machine; continued PCB soldering, wiring harness construction and motor lead/connector work.	Software reflected in `Progress.md`; physical work reconstructed.
9	2025-12-01 to 2025-12-07	Jump behaviour and physical iteration	Implemented jump preparation task; revised printed parts and mechanical mounting as needed; continued checking harness strain relief and motor wiring reliability.	Software reflected in `Progress.md`; physical work reconstructed.
10	2025-12-08 to 2025-12-14	Xbox BLE controller	Integrated NimBLE-Arduino; added Xbox controller scan/connect, button mappings and 50 Hz BLE processing task; tested manual input mapping.	Reflected in `Progress.md` 2025/12/8-12/14.
11	2025-12-15 to 2025-12-21	Christmas period - reduced work	Added motor calibration, NVS persistence, IMU debug commands, ADS1115 voltage monitoring and battery compensation; light hardware cleanup.	Christmas vacation started; reduced but non-zero work.
12	2025-12-22 to 2025-12-28	Christmas period - command system	Implemented UART2 command receiving and queue system; added movement/action commands, queue control and composite command parsing.	Reduced development period.
13	2025-12-29 to 2026-01-04	Christmas period - cleanup	Continued command queue cleanup, documentation and debugging notes; prepared to pause major development for exams.	Reduced development period.
14	2026-01-05 to 2026-01-11	Exam period	Minimal project development; priority shifted to exam revision. Reviewed design notes only where possible.	Exam season.
15	2026-01-12 to 2026-01-18	Exam period	Minimal project development; no major implementation milestone.	Exam season.
16	2026-01-19 to 2026-01-25	Exam period	Minimal project development; maintained project notes and post-exam restart plan.	Exam season.
17	2026-01-26 to 2026-02-01	Restart after exams	Resumed development; planned cross-step and input arbitration changes; rechecked hardware/firmware state after break.	Transition week.
18	2026-02-02 to 2026-02-08	Cross-step and kinematics	Added cross-step walking and BLE input enable/disable; generated MATLAB kinematics functions for leg position, speed, VMC conversion and LQR gains.	Reflected in `Progress.md` 2026/2/1-2/10.
19	2026-02-09 to 2026-02-15	LQR/VMC integration	Built leg pose update task; defined LQR state vector; integrated main `CtrlBasic_Task` at 4 ms; added wheel output = LQR + yaw PID.	Reflected in `Progress.md`.
20	2026-02-16 to 2026-02-22	Debugging and diagnosis	Reworked triggers for gentle stand-up; added leg motor test mode; added IMU serial debug; investigated ToF/OpenMV/host architecture; designed/printed sensor mounting base.	Reflected in `Progress.md` 2026/2/15-2/22.
21	2026-02-23 to 2026-03-01	Mechanical/electrical consolidation	Continued physical integration, printed part modification, PCB/wiring cleanup, harness reliability checks and motor soldering/connector improvements; continued control tuning.	Reconstructed physical build work not fully captured in git.
22	2026-03-02 to 2026-03-08	Hardware reliability and tuning	Iterated on mechanical fit and wiring strain relief; checked power distribution and sensor mounting; continued balance tuning and manual control testing.	Reconstructed.
23	2026-03-09 to 2026-03-15	Integration testing and early evidence planning	Combined embedded control, manual control, wiring and mechanical build into longer bench tests; started the early experiment matrix and identified what data would be needed for control, safety and teleoperation evidence.	Reconstructed; marks the start of Testing/Data/Report activity.
24	2026-03-16 to 2026-03-22	Test plan, design review and scope control	Reviewed navigation/vision ambitions against time and risk; began narrowing scope toward safe teleoperation and measurable evidence rather than full autonomy; prepared initial data-template and report-outline planning.	Reconstructed; aligns with later architecture decision.
25	2026-03-23 to 2026-03-29	Pre-Easter report and data consolidation	Prepared report outline and experiment ideas; continued hardware cleanup; identified need for latency, watchdog and repeatability evidence; organised the project evidence into control, communication/safety and vision categories.	Reconstructed.
26	2026-03-30 to 2026-04-05	Easter vacation	Reduced formal schedule; used time for catch-up, report planning, hardware review and software architecture consolidation.	Easter vacation.

Week	Dates	Main focus	Activities and outputs	Notes / status
27	2026-04-06 to 2026-04-12	Easter vacation	Continued catch-up; refined final-report positioning around local embedded balance plus supervisory teleoperation; prepared for ROS/camera bring-up.	Easter vacation.
28	2026-04-13 to 2026-04-19	Easter vacation / restart	Finalised plan to focus on measurable evidence; prepared experiment matrix, report outline and camera/host software route.	Easter vacation ended 2026-04-18.
29	2026-04-20 to 2026-04-25	Final integration, testing and report sprint	Brought up the ROS 2 WiFi camera module path; implemented ROS 2 `wheeleg_vision_bridge`; added WiFi TCP command safety; collected E4/E5/E6/E8/E10/E11 data; collected and organised E1/E2/E3/E4b/E9 physical datasets; drafted Chapters 1-5 and references.	Reflected heavily in `Progress.md` and experiment data.
30	2026-04-26 to 2026-05-01	Final measured-data integration and submission preparation	Replaced earlier analysis scaffolding with measured E1-E11 evidence, rebuilt the LaTeX report, checked appendix consistency, and prepared the public repository evidence for marking.	Final report sign-off window.

Key Schedule Deviations and Responses

Deviation	Cause	Response	Effect on final report
More time spent on low-level balance control than planned	Wheel-legged dynamics, motor wiring and IMU/control tuning were more difficult than early planning assumed	Prioritised local ESP32 stabilisation and reduced ambition for full autonomous navigation	Strengthened the final architecture argument around local real-time control
Physical build work was larger than software commits suggest	CAD, printing, PCB work, soldering and wiring harness construction happened outside git	Reconstructed the project management record and linked physical work to integration risks	Appendix B explains non-git effort and manufacturing workload
ROS 2 was scoped away from low-level control	WiFi/ROS/camera timing is not deterministic enough for a 4 ms balance loop	Used ROS 2/MediaPipe only for supervisory teleoperation	Became a central design choice tested by E5/E10/E11
Exam and vacation periods reduced continuity	Christmas, exam season and Easter break interrupted continuous lab work	Used reduced periods for documentation, planning, light cleanup and later sprint work	Project plan records realistic schedule constraints
Some physical experiment data was collected late in the reporting window	Full robot hardware/data availability lagged behind software and report timeline	Prioritised the final hardware session around E1/E2/E3/E4b/E9 and replaced earlier analysis scaffolding with measured datasets	Final report tables use measured E1-E11 evidence, with raw CSV files retained in Appendix G